

FIM24725

Micro Integrated Coherent Receiver

(Rev. A)

March 18th, 2019
Customer: Standard

1. GENERAL DESCRIPTION

- Small size: 23 mm x 12 mm x 5.5 mm
- PIN/TIA+AGCA(*1), photo receiver, PBS,VOA(*2), MPD and Optical 90 degree hybrid in one package
- Power consumption: < 1.7 W
- Co-Planer Waveguide (CPW) and SMT type RF interface
- OIF compliant

Note 1: Auto Gain Control Amplifier

Note 2: Normally bright Type



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2. PART NUMBER

FIM24725

3. SPECIFICATIONS

3.1 ABSOLUTE MAXIMUM RATINGS

	Parameter	Condition	Min.	Max.	Unit
1	Storage Temperature	Ambient	-40	85	degC
2	Storage Humidity (*1)	No condensation	5	95	%RH
3	Optical Input Power	Peak, Sig or LO	-	17	dBm
4	Photo Diode Reverse Voltage	Signal (*2)	-0.8	7	V
		Monitor	-0.5	20	V
5	Amplifier Supply Voltage		-0.5	4.5	V
6	Amplifier Control Voltages		-0.5	4.5	V
7	VOA Supply Voltage		0	5.2	V
8	Electro Static Discharge	Human Body Model	-250	250	V
9	Fiber Bend Radius		12	-	mm

Note 1: Telcordia GR-63-CORE compliant (Not to exceed 0.024 kg water/kg of dry air).

Note 2: VPD_*** in "APPENDIX 3: PIN FUNCTIONS". Without amplifier supply voltage applied.

3.2 OPERATING CONDITION

	Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
1	Operating Temperature			-5	-	85	degC
2	Operating Humidity (*1)			5		85	%RH
3	Frequency Range (Wavelength Range)			191.35 (1566.72)	-	196.20 (1527.99)	THz (nm)
4	Signal Power: Operating Range	P(s)	Average	-18	-10	0	dBm
5	Local Oscillator Input Power	P(Lo)		3	-	13	dBm
6	Photodiode Supply Voltage			4.75	5.0	5.25	V
7	Amplifier Supply Voltage	Vcc		3.135	3.3	3.465	V
8	Amplifier Supply Current	Icc	Total of TIAs	-	360	400	mA
9	Gain adjustment voltage	GC		0	-	Vcc	V
10	Output voltage swing adjustment	OA		0	-	Vcc	V
11	Output shutdown	SHD	Enable	0	-	0.8	V
12	(*2)		Disable	2	FLT (*3)	Vcc	
13	Gain control mode	MC	MGC	0	-	0.8	V
14	selection (*3)		AGC	2	FLT (*3)	Vcc	
15	VOA adjustment voltage			0	-	4.8	V
16	VOA current consumption			-	-	80	mA
17	Power consumption			-	-	1.7	W

Note 1: Temperature and relative humidity limits are in compliance with Telcordia GR-63-CORE.

Note 2: Digital control

Note 3: Floating condition

3.3 OPTICAL AND ELECTRICAL SPECIFICATIONS (Target)

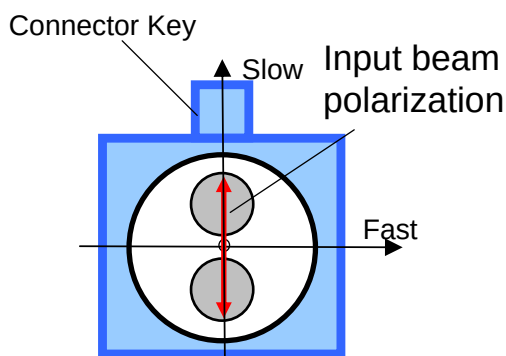
	Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
1	Symbol Rate			-	-	32	GBaud
2	Output swing adjustment range (*1)		Peak to peak, differential, AC coupled	300	-	900	mVppd
3	Total Harmonic Distortion	THD		-	-	5	%
4	Common Mode Rejection Ratio: DC, Signal to I & Q DC, LO to I & Q 22GHz, Signal to I & Q 22GHz, LO to I & Q	CMRR		-	-	-20 -12 -16 -10	dBe
5	Small signal Bandwidth (3dB)			-	22	-	GHz
6	Low Frequency Cutoff		AC coupling	-	-	100	kHz
7	Phase Error		Between XI and XQ and between YI and YQ	-5	-	5	deg.
8	Optical Reflectance		Signal and LO ports Per ITU-T G.959.1	-	-	-27	dB
9	Output Electrical Return Loss (S22): f < 16 GHz 16 GHz < f < 24 GHz 24 GHz < f < 32 GHz			- - -	- - -	-10 -8 -6	dB dB dB
10	Skew: p, n					2	ps
11	Channel skew		(*2)	-	-	10	ps
12	Channel skew variation		(*3)	-	-	5	ps
13	Peak Detection Voltage			0	-	2	V
14	Maximum Differential Transimpedance		MGC	3200	-	-	Ω
	Minimum Differential Transimpedance		MGC	-	-	250	
15	Maximum Gain Control Bandwidth			-	5	-	MHz
16	Responsivity	Rsig	(*4)	0.025	0.05	-	A/W
17	Responsivity difference		Signal, between p and n	-	-	1	dB
18	Polarization extinction ratio			18		-	dB
19	Photodiode Dark Current			-	1	-	nA
20	Maximum VOA attenuation range	IL		10	-	-	dB
21	VOA Response Time	(*5)	0 < IL ≤ 10dB	-	-	20	ms
22	Monitor PD responsivity	Rmon			0.05		A/W
23	Monitor PD to LO input optical isolation(*6)				45		dB

- Note 1: Output swing can be controlled by AGC or MGC between the min. value and the max. value. This does not mean output swing can be controlled whole range from the min. value to the max. value.
- Note 2: Time difference between earliest and latest channel. Includes channel skew variation.
- Note 3: Temporal variation in the skew between any 2 channels due to case temperature, wavelength, input optical power, amplifier gain, and aging. Time for channel defined as mean of p and n.
- Note 4: For responsivity from signal and LO port, linearly polarized optical input is aligned 45 degree to PBS and slow axis of PMF, respectively. VOA attenuation is 0dB.
- Note 5: 90% of rise / fall time of step response for VOA adjustment voltage.

3.4 MECHANICAL SPECIFICATIONS

	Parameter		Condition	Spec	Unit
1	Mechanical Dimensions			23 x 12 x 5.5	mm
2	Fiber	Type		Sig : SMF LO : PMF	-
3		Mode field Diameter	SMF28 or its complaint	10	μm
4		Flammability	-	Signal : UL94-V0 LO : Oxygen index is about 18%	-
5		Color	-	Sig: Yellow (*2) LO: Transparent	-
6		Diameter	-	Sig : 0.5 typical LO : 0.4 typical	mm
7	Fiber length		-	>= 1000	mm
8	Optical Connector		-	Sig : LC LO : LC (*1)	-
9	Optical end face		-	SPC	-

Note 1 : Polarization of the LO input to be aligned to the slow-axis. The connector key is adjusted to the direction within +/-5degree from slow-axis.



3.5 DIMENSIONS

Appendix 2 shows the dimensions of the receiver.

3.6 PIN FUNCTIONS

Appendix 3 shows the pin functions of the receiver.

4. MARKING

The following information shall be marked on the receiver.

- (1) Symbol Mark: 
- (2) Part Number: FIM24725
- (3) Serial Number : BBBAAAAA
 - BBB: Keyword (3 digits of alphabets or figures)
 - AAAAA: Serial Number (5 digits of symbols in 34-base (0~9, A~Z except for "I" and "O"). Reset every week.)
- (4) Country of manufacture: MADE IN xxxxx (country name)

5. PACKING

Each Receiver shall be individually packaged in a shipping container to prevent damage that may result from pressure, vibration or movement during shipment.

The following information shall be marked on the shipping container.

- (1) Symbol Mark
- (2) Part Number
- (3) Serial Number

6. HANDLING PRECAUTION

- (1) When handling the Receiver, hold the package part, not the fiber.
- (2) The attached fiber should be handled very carefully. Do not twist nor exceed a pull force greater than 500gf. A handling radius should not be less than 15mm.
- (3) This receiver is susceptible to damage as a result of electrostatic discharge (ESD). Appropriate handling precautions against electrostatic discharge must be taken during handling and testing.
- (4) The receiver includes precision optics. If package of the receiver is deformed by excess load, the receiver may be degraded in its characteristics or damaged.
- (5) In order to prevent the degradation or damage, the receiver should be mounted on flat area (flatness < 15um, surface roughness < 12.5um) larger than package bottom area including package flanges by M1.2 screws with washer (necessary), spring washer (recommended) and 0.376kgf*cm torque.

7. TEST DATA

Each receiver shall be supplied with the following test data.

No	Item	Condition
1	Responsivity from Sig port	-5degC, 1568nm
2	Responsivity from LO port	-5degC, 1568nm
3	Optical return loss in Sig port	-5degC, 1568nm
4	Optical return loss in LO port	-5degC, 1568nm
5	Amplifier power supply current	85degC
6	VOA adjustment voltage of 10dB attenuation	-5degC, 1568nm

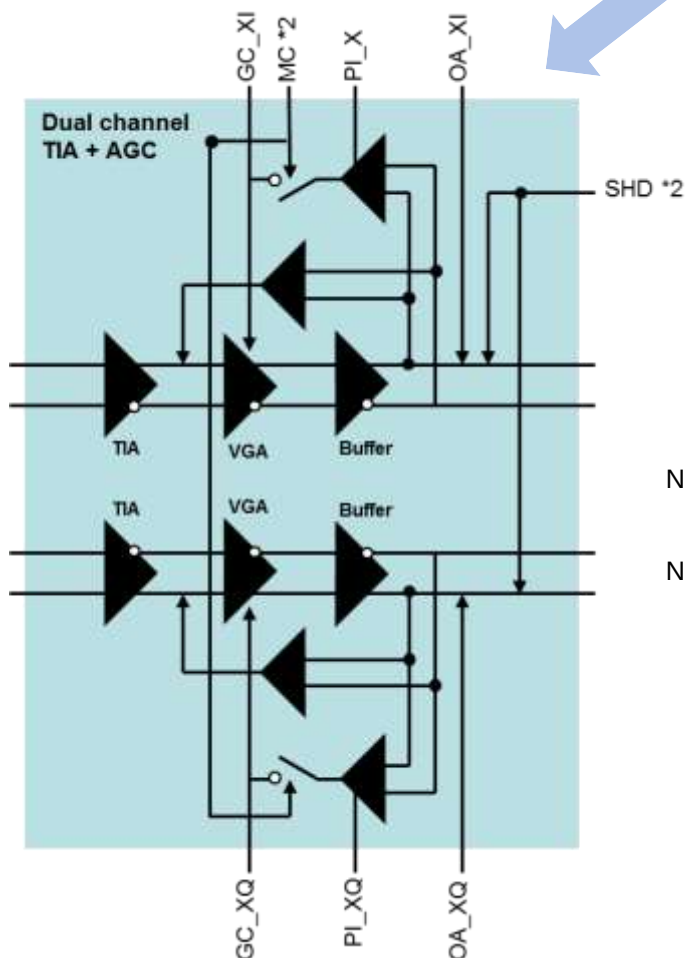
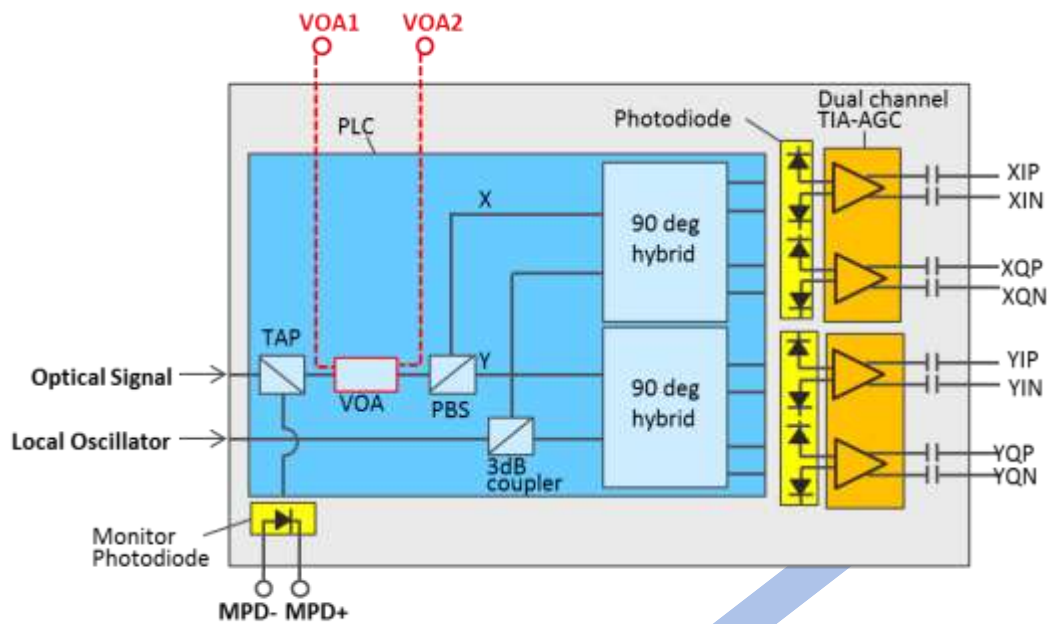
8. SAFTY PRECAUTION

This Product is designed, developed and manufactured as contemplated for general use, including without limitation, general office use, personal use, household use, and ordinary industrial use, but is not designed, developed and manufactured as contemplated for use accompanying fatal risks or dangers that, unless extremely high safety is secured, could lead directly to death, personal injury, severe physical damage or other loss (hereinafter "High Safety Required Use"), including without limitation, nuclear reaction control in nuclear facility, aircraft flight control, air traffic control, mass transport control, medical life support system, missile launch control in weapon system. You shall not use this Product without securing the sufficient safety required for the High Safety Required Use. If you wish to use this Product for High Safety Required Use, please consult with our sales representatives in charge before such use.

9. Export notes

Exportation/release of this product may require necessary procedures in accordance with the regulations of the Foreign Exchange and Foreign Trade Control Law of any applicable export control.

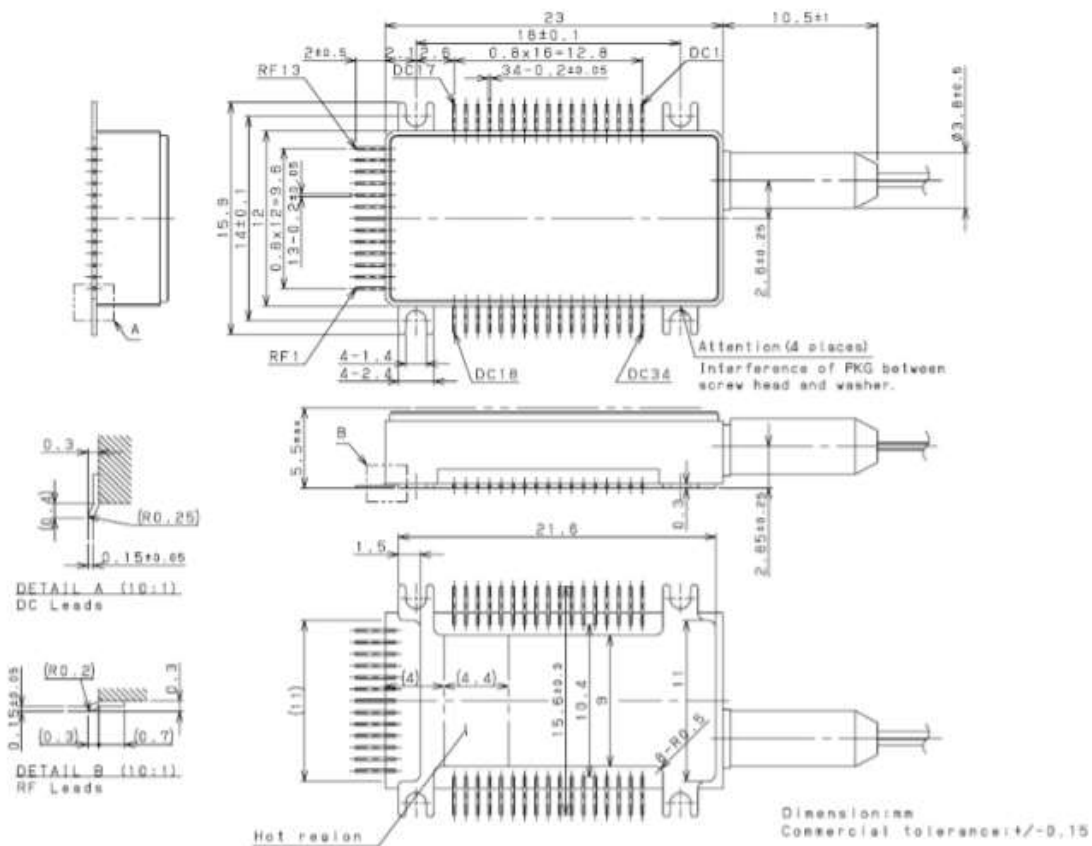
APPENDIX 1: BLOCK DIAGRAM



Note 1: This diagram represents X channel. Y channel has same diagram.

Note 2: This function controls XI, XQ, YI and YQ channels simultaneously.

APPENDIX 2: DIMENSIONS



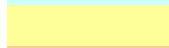
APPENDIX 3: PIN FUNCTIONS

DC pins

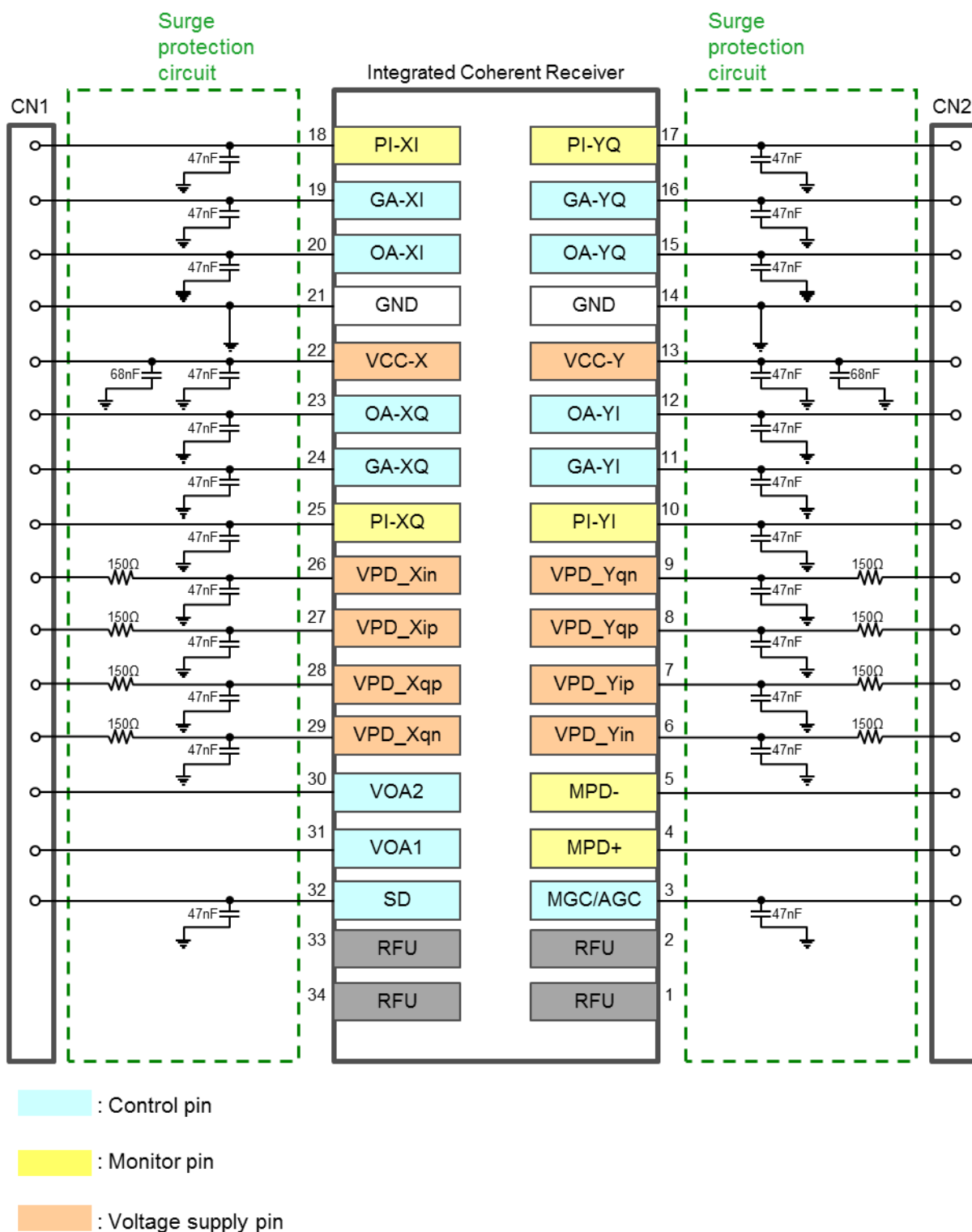
No	NAME	FUNCTION
1	RFU	Reserved for Future Use
2	RFU	Reserved for Future Use
3	MC/AGC	MGC/AGC Selection
4	MPD+	Monitor Photodiode Cathode
5	MPD-	Monitor Photodiode Anode
6	VPD_YIn	Photodiode power supply (YIn)
7	VPD_YIp	Photodiode power supply (YIp)
8	VPD_YQp	Photodiode power supply (YQp)
9	VPD_YQn	Photodiode power supply (YQn)
10	PI-YI	Peak Indicator YI
11	GA-YI	Gain Adjust YI
12	OA-YI	Output Amplitude Adjust YI
13	VCC_Y	Amplifier power supply (YI and YQ)
14	GND	Ground
15	OA-YQ	Output Amplitude Adjust YQ
16	GA-YQ	Gain Adjust YQ
17	PI-YQ	Peak Indicator YQ
18	PI-XI	Peak Indicator XI
19	GA-XI	Gain Adjust XI
20	OA-XI	Output Amplitude Adjust XI
21	GND	Ground
22	VCC_X	Amplifier power supply (XI and XQ)
23	OA-XQ	Output Amplitude Adjust XQ
24	GA_XQ	Gain Adjust XQ
25	PI-XQ	Peak Indicator XQ
26	VPD_XIn	Photodiode power supply (XIn)
27	VPD_XIp	Photodiode power supply (XIp)
28	VPD_XQp	Photodiode power supply (XQp)
29	VPD_XQn	Photodiode power supply (XQn)
30	VOA2	VOA GND
31	VOA1	VOA control
32	SD	Shutdown
33	RFU	Reserved for Future Use
34	RFU	Reserved for Future Use

RF pins

No	Name
1	GND
2	XIp
3	XIn
4	GND
5	XQp
6	XQn
7	GND
8	YIp
9	YIn
10	GND
11	YQp
12	YQn
13	GND

 : Control pins
 : Monitor pins
 : Power supply pins

APPENDIX 4: Pin functions and typical circuit configuration

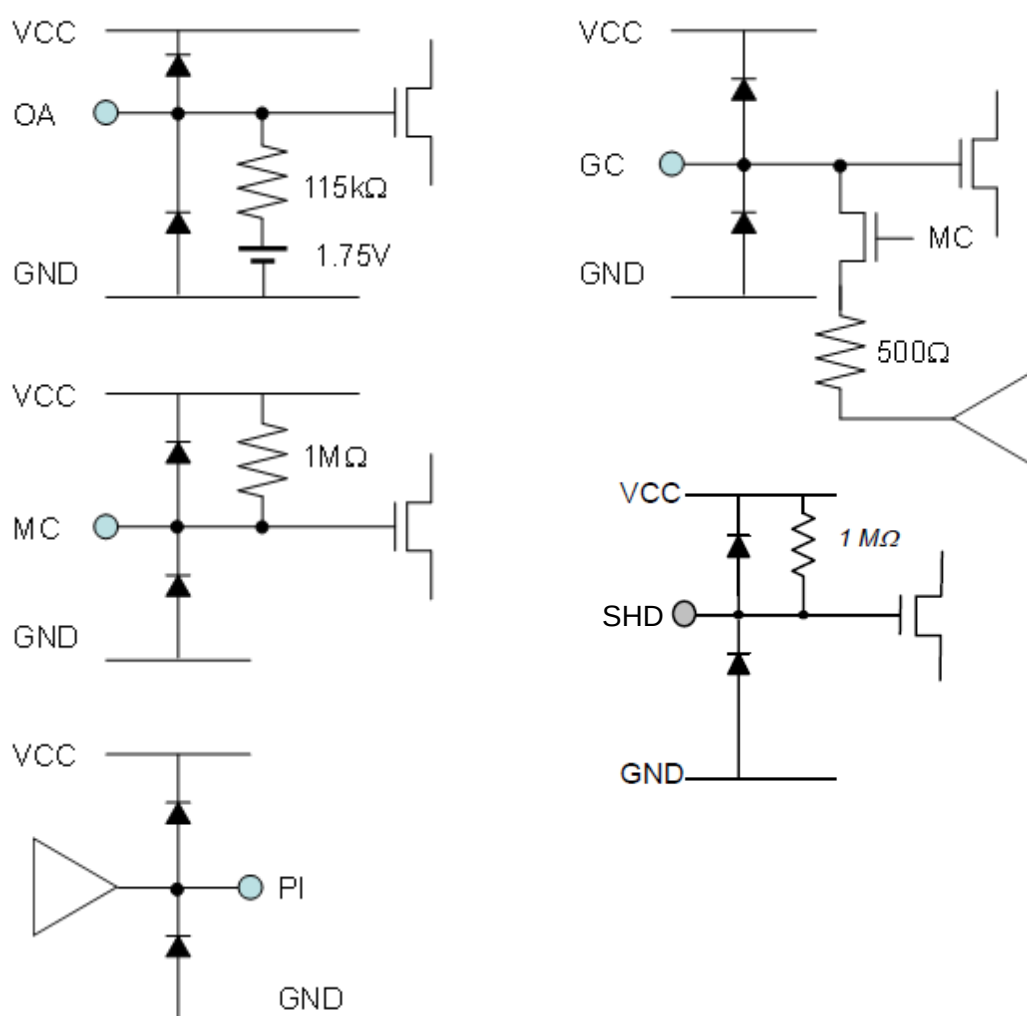


APPENDIX 5: DEFINITION OF SPECIFICATIONS

1. CHANNEL NAME

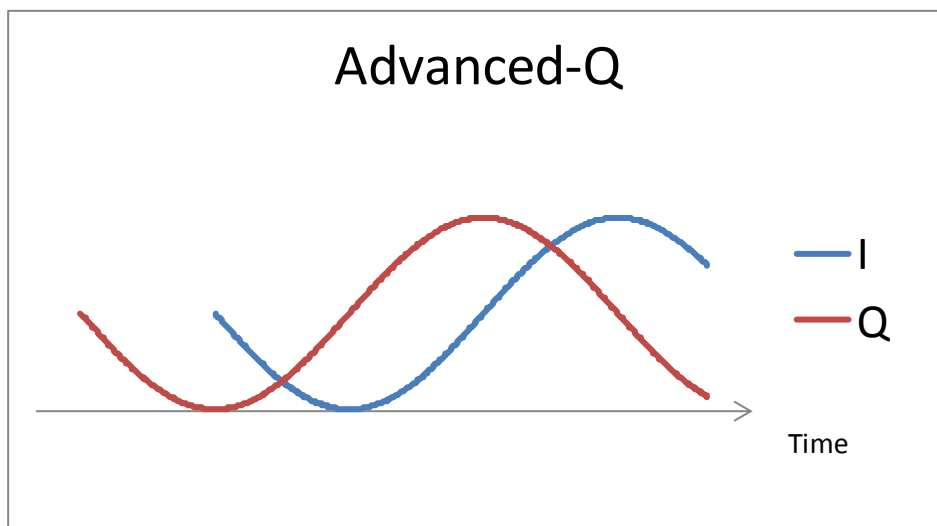
CHANNEL NAME	DEFINITION
X and Y	Nominally orthogonal polarizations (no absolute polarization state implied)
I and Q	Nominally orthogonal phases
p and n	Alternate polarities for the differential signals (no absolute polarities or sign implied).

APPENDIX 6: EQUIVALENT CIRCUITS OF CONTROL PINS



APPENDIX 7: I/Q phase relation

Figure below shows I/Q phase relation when optical frequency of the signal is higher than that of the local oscillator.



CHANGE NOTES FOR SPECIFICATION

Rev.	Date	Description
A	March 18 th , 2019	Generated specification

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